T-1: A shared spectral-action normalization test

Exact internal sums, a conditional radius, and a same-moment counterexample

Research calculation prepared for Christopher Michael Baird with ChatGPT

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Result and model identity

A newly stipulated common bosonic spectral action supplies both a gravitational kinetic term and a hypercharge kinetic term from the same operators. On the inherited four-copy monopole sphere, it gives a calculable **relation** between those normalizations. For an exponential spectral profile and one explicitly allowed cosmological subtraction, the round-product ansatz has a stationary radius

$$\Lambda r_* = 1.30078328322736\ldots$$

with positive curvature of its Einstein-frame radius potential. Its normalizations obey

$$g_Y^{-2} = 0.330363122965351\ldots\,\kappa,$$

$$M_4 r_* = 0.801195658830816\ldots\sqrt{\kappa},$$

and therefore

$$\boxed{\frac{g_Y^{-2}}{(M_4 r_*)^2} = 0.514652856944484\ldots.}$$

These are conditional results for a **new bosonic action postulate**, not a further one-loop correction to the old Einstein–Maxwell theory, and not a numerical prediction of the measured fine-structure constant. No measured coupling is used.

The positive trace alone has positive vacuum energy on this ansatz and cannot provide its Minkowski reference vacuum without additional contributions. The cosmological subtraction used to obtain the stationary radius is an explicit extra local operator, not a result of spectral regularization.

More importantly, two strictly positive spectral profiles with identical normalization and all moments multiplying the leading six-dimensional coefficients through a_6 give different exact compactified couplings. Fixing those leading spectral data does not uniquely fix the gauge strength. The overall coefficient κ, the physical spectral profile, the quantum definition of the postulate, and electromagnetic embedding remain additional requirements.

The audit performs **32 symbolic identities and 27 numerical checks**. The immediately preceding saved spectral-matching script was rerun: its **54 checks pass**. No historical boundary-value campaign or global anomaly proof was rerun in this pass.

The supplied corpus distinguishes original actions from illustrative compactifications and treats the electromagnetic normalization bridge as unresolved [P1]. Its audit inserts also distinguish historical claims from the current technical statement [P2]. This note preserves those boundaries.

1. What changes, and what is retained

The background is the unbroken product of a four-dimensional spacetime and a round sphere, with unit X monopole flux and no internal hypercharge or color flux. The fermions are the same four copies of the repaired six-dimensional Weyl block used in the preceding calculation [P3]. Their minimal operators contain no bulk masses or Higgs expectation values.

Field	Color dimension	x	y	6D chirality
Q	3	2	$1/3$	$+$
L	1	2	-1	$+$
u	3	1	$4/3$	$-$
d	3	1	$-2/3$	$-$
e	1	1	-2	$-$
spectator	3	0	$1/3$	$+$

Hypercharge uses

$$D_\mu = \partial_\mu - i(y/2)\mathscr{B}_\mu + \cdots .$$

The weak group in this comparison remains geometric; it is not an extra six-dimensional weak multiplicity in the table. It is not included as an external background in the calculation below.

The change is the bosonic matching postulate

$$\boxed{S_{\rm sp} = \frac{\kappa}{2} \sum_{i\in\text{four copied blocks}} \mathrm{Tr}_{\rm Dirac}\ f(D_i^2/\Lambda^2), \qquad \kappa > 0.}$$

The color representation is included in each trace. The factor $1/2$ is an explicitly chosen parity-even Weyl normalization using the associated full Dirac operator. It does not compute the anomalous phase of a chiral determinant. It does not add physical mirror fermions.

The positive, rapidly decreasing profile f, the spectral energy scale Λ, and the common normalization κ are inputs. A spectral action as a physical bosonic functional is an established proposal [1]. It is conceptually different from a regulator used to define $-\log\det D$. The previous loop thresholds must not be counted as the same functional a second time.

In the first comparison, the ordinary independent Einstein and Maxwell coefficients are replaced by coefficients derived from $S_{\rm sp}$. Other hypothetical tensor/anomaly sectors are not reconstructed by this trace. Their global consistency remains conditional. Nonlocality, Lorentzian continuation, reflection positivity, and radiative stability of the common-action restriction are not established here.

2. Exact internal heat sums and external normalization

For nonnegative monopole index n, the massive internal modes have

$$m_{p,n}^2 = \frac{p(p+n)}{r^2}, \qquad d_{p,n} = 2p+n, \qquad p \geq 1.$$

There are also n chiral zero modes. The monopole and ordinary sphere spectra are standard inputs [3,4]. Unlike the preceding heavy-only determinant, the spectral action counts those zero modes: it is a trace of the full stipulated operator, not an inverse-mass expansion for integrating them out.

Set $R = \Lambda r$ and define

$$h_n(R) = \frac{n}{2} + \sum_{p=1}^{\infty}(2p+n)e^{-p(p+n)/R^2}.$$

The half-integer zero-mode factor is the parity-even normalization of four-dimensional Weyl versus Dirac modes. The exact multiplicities are

$$D_0 = 12, \quad D_1 = 28, \quad D_2 = 16,$$

and the hypercharge-square weights are

$$w_0 = \frac{1}{3}, \quad w_1 = \frac{32}{3}, \quad w_2 = \frac{4}{3}.$$

Thus

$$H(R) = 12h_0 + 28h_1 + 16h_2,$$

$$H_Y(R) = \frac{1}{3}h_0 + \frac{32}{3}h_1 + \frac{4}{3}h_2.$$

For $f(s) = e^{-s}$, separation of the product Dirac operator gives the external four-dimensional heat trace multiplied by these internal sums. The standard Dirac heat coefficients contain

$$\operatorname{tr} a_0 = 4, \qquad \operatorname{tr} a_2 = -\frac{1}{3}\mathcal{R}_4, \qquad \operatorname{tr} a_4|_{\mathscr{F}^2} = \frac{2}{3}q^2\mathscr{F}^2,$$

with the overall $(4\pi s)^{-2}$ suppressed. The a_2 sign follows from $E = -\mathcal{R}/4$, and the a_4 coefficient combines $E^2/2$ and $\Omega_{\mu\nu}\Omega^{\mu\nu}/12$ [2]. These produce the Euclidean low-external-derivative action

$$S_{4,E} = \int \sqrt{g}\left[U_{\rm sp}(R) - \frac{M_4^2(R)}{2}\mathcal{R}_4 + \frac{Z_Y(R)}{4}\mathscr{F}^2 + \cdots\right],$$

where

$$\boxed{U_{\rm sp}(R) = \frac{\kappa\Lambda^4}{4\pi^2}H(R),}$$

$$\boxed{M_4^2(R) = \frac{\kappa\Lambda^2}{24\pi^2}H(R), \qquad Z_Y(R) = g_Y^{-2} = \frac{\kappa}{6\pi^2}H_Y(R).}$$

The sums are exact in internal curvature/flux on the reference product. The expansion is still in external derivatives and external field strengths. It requires external momenta and curvatures small compared with the relevant spectral and compactification scales. It is not an exact action for arbitrary six-dimensional fluctuations.

For this exponential profile, the relative normalization is fixed at any specified R:

$$\boxed{\frac{\Lambda^2 Z_Y(R)}{M_4^2(R)} = \frac{4H_Y(R)}{H(R)}.}$$

This is an actual shared-action relation. It replaces two independent coefficients only because the shared-action rule was added as a physical assumption.

At large R,

$$H(R) = 56R^2 - \frac{28}{3} + \frac{101}{15R^2} + O(R^{-4}),$$

$$H_Y(R) = \frac{37}{3}R^2 - \frac{37}{18} + \frac{203}{180R^2} + O(R^{-4}).$$

The first common-action ratio tends to $37/42$, not to a putative electromagnetic inverse fine-structure constant. At the root below R is near one, so a large-R truncation is not used. Exact spectral-action computations on symmetric spaces provide a standard alternative to relying only on the asymptotic heat series [5].

3. Vacuum energy: a separate necessary condition

Every $h_n(R)$ is positive. In particular, the massless sectors give

$$H(R) \geq 28/2 + 16 = 30.$$

Hence $U_{\rm sp}(R) > 0$ at every finite R. For constant internal fields and a flat external metric, the zero-derivative external Einstein equation requires the total four-dimensional vacuum energy to vanish. The unmodified positive exponential trace cannot meet that requirement.

This is not a general prohibition on spectral-action models: additional sectors, vacuum subtractions, other physical action terms, or different external geometry change the problem. It is an exact obstruction to this restricted trace-only Minkowski ansatz.

To test whether the normalization relation can coexist with a stationary radius, explicitly allow one additional six-dimensional cosmological density,

$$\Delta S_{6,E} = -\frac{\kappa c \Lambda^6}{16\pi^3} \int \sqrt{g_6}.$$

On the product it gives

$$U_J(R) = \frac{\kappa \Lambda^4}{4\pi^2} [H(R) - cR^2].$$

No independent Maxwell or Einstein coefficient is reintroduced in this comparison. Nevertheless, c is a new local parameter. Demanding a flat stationary solution fixes it by an imposed vacuum condition; this is not a solution of the cosmological constant problem.

The two conditions are

$$H(R_*) - cR_*^2 = 0, \qquad H'(R_*) - 2cR_* = 0.$$

Eliminating c gives a one-variable equation:

$$\boxed{R_* H'(R_*) - 2H(R_*) = 0, \qquad c = \frac{H(R_*)}{R_*^2}.}$$

The computed root and associated quantities are

$$R_* = 1.3007832832273603186\ldots,$$

$$H_* = 89.862452687027315712\ldots, \quad H_{Y,*} = 19.563319994258713198\ldots,$$

$$c_* = 53.109030549995206235\ldots.$$

The radius dependence of the gravitational coefficient must be included in the Einstein-frame potential. Fixing its normalization at the stationary point gives

$$U_E(R) = \frac{\kappa\Lambda^4}{4\pi^2}\left[\frac{H_*}{H(R)}\right]^2 [H(R) - c_* R^2].$$

Because the bracket and its first derivative vanish at R_*,

$$\frac{4\pi^2}{\kappa\Lambda^4}\frac{d^2U_E}{dR^2}\bigg|_* = H''(R_*) - 2c_* = 25.078214769770544259\ldots > 0.$$

This establishes a local minimum of the restricted radius potential. The radion derivative terms and canonical kinetic normalization have not been calculated for the exact spectral action. A positive potential curvature is therefore not yet a physical radion mass or a full stability theorem. No claim of a unique root over all possible profiles or topologies is made.

The resulting hypercharge and gravitational relation is

$$g_Y^{-2} = 0.33036312296535089940\ldots\,\kappa,$$

$$M_4 r_* = 0.80119565883081607604\ldots\sqrt{\kappa}.$$

Equivalently,

$$\boxed{g_Y^{-2} = 0.51465285694448366276\ldots\,(M_4 r_*)^2.}$$

It is a conditional relation between a gauge strength and the compactification scale relative to gravity, not an absolute coupling. Even for fixed exponential profile, changing κ changes g_Y. A common bosonic normalization is genuine input; its cancellation from one ratio does not prove it disappears from all observables.

4. Two positive profiles with the same leading local data

One might stipulate $\kappa = 1$ and hope the leading spectral moments select a unique value. The following explicit counterexample tests that hope without changing any fermion representation, monopole index, or charge normalization.

Take positive exponential mixtures

$$f(u) = \sum_j w_j e^{-a_j u}, \qquad w_j > 0, \quad a_j > 0.$$

Define

$$P_f(R) = \sum_j \frac{w_j}{a_j^2} H(R/\sqrt{a_j}),$$

$$J_f(R) = \sum_j \frac{w_j}{a_j} H(R/\sqrt{a_j}),$$

$$T_f(R) = \sum_j w_j H_Y(R/\sqrt{a_j}).$$

Linearity of the trace gives

$$U_J = \frac{\kappa\Lambda^4}{4\pi^2}(P_f - cR^2), \quad M_4^2 = \frac{\kappa\Lambda^2}{24\pi^2} J_f, \quad Z_Y = \frac{\kappa}{6\pi^2} T_f.$$

Flat stationary radii solve $RP_f' - 2P_f = 0$. At such a point the Einstein-frame potential curvature has the sign of $P_f'' - 2c$.

Choose

$$a_j = \frac{5269}{8220} j, \qquad j = 1, 2, 3, 4, 5,$$

$$v = (1, -32, 162, -256, 125),$$

$$\boxed{w_j^{\pm} = \frac{1}{5} \pm \frac{v_j}{2560}, \qquad f_{\pm}(u) = \sum_{j=1}^{5} w_j^{\pm} e^{-a_j u}.}$$

All ten weights are strictly positive. The key integer identities are

$$\boxed{\sum_{j=1}^{5} \frac{v_j}{j^k} = 0, \qquad k = 0, 1, 2, 3.}$$

With moments

$$\mathfrak{m}_k[f] = \frac{1}{\Gamma(k)} \int_0^{\infty} u^{k-1} f(u)\, du,$$

both profiles have

$$f_+(0) = f_-(0) = 1,$$

$$\mathfrak{m}_1[f_{\pm}] = \mathfrak{m}_2[f_{\pm}] = \frac{18769}{26345},$$

$$\mathfrak{m}_3[f_{\pm}] = \frac{658531217359}{731399400545}.$$

Thus they match the coefficients multiplying the six-dimensional heat invariants a_0, a_2, a_4, a_6. In particular, the volume, Einstein, gauge, and zero-power heat data match. The scale factor fixes $\mathfrak{m}_2/\mathfrak{m}_1 = 1$, so the difference is not simply a rescaling of Λ.

Nevertheless, the exact internal sums yield:

Quantity	f_-	f_+
Stationary Λr_*	1.30211764936900	1.30187576169391
Required c_*	48.3989788902990	48.3978806753453
Positive potential curvature factor	16.0006637517561	16.0655185006695
g_Y^{-2}/κ	0.245345972549864	0.245467728342579
$M_4 r_*/\sqrt{\kappa}$	0.685557884695684	0.685267055890428
$g_Y^{-2}/(M_4 r_*)^2$	0.522024243990561	0.522726714970681

The two inverse hypercharge strengths differ by

$$\frac{Z_{Y,+} - Z_{Y,-}}{\kappa} = 0.00012175579271525909\ldots,$$

or about 0.0496262% relative to the minus profile. That percentage is a comparison between two defined mathematical models, not an experimental discrepancy.

Both examples impose the same flat-vacuum rule, which determines their slightly different cosmological subtractions. Both have positive restricted radial potential curvature. Neither is asserted to be the unique or observed vacuum.

As a control, replacing the pure exponential by e^{-3u} and rescaling the root by $\sqrt{3}$ leaves g_Y^{-2}/κ and $M_4 r/\sqrt{\kappa}$ unchanged. The deformations above do change those quantities. They are genuinely different physical profiles, not that elementary scale redefinition.

When the profile is part of the postulated physical action, this is a difference between theories. It is not a subtraction-scheme change, and it cannot be canceled by adjusting a regulator convention while claiming to keep the same theory.

5. Reproduction and conservative precision

The exact tests check the inherited field multiplicities and charge traces, external Dirac heat coefficients, the gravitational/gauge ratio, zero-mode bookkeeping, the internal asymptotics, and the profile-moment identities.

The numerical tests use 65-decimal working precision, analytic first and second derivatives of the heat sums, a second independent floating-point root solver, and internal truncations at 40 and 60 radial levels. At each displayed root, the code verifies the zero-energy condition, radial stationarity, and the Hessian after the radius-dependent Einstein-frame rescaling. The reported long decimal strings are reproducibility data, not certified enclosures of exact real roots. The shorter tables should be used for ordinary reporting.

The heat-sum tails have a simple analytic bound. Once the summand is decreasing for $p \geq N$,

$$\sum_{p>N} (2p+n) e^{-p(p+n)/z^2} \leq \int_N^\infty (2x+n) e^{-x(x+n)/z^2}\, dx = z^2 e^{-N(N+n)/z^2}.$$

At the conservative upper radius $R = 1.4$, the resulting value bounds for P_f, J_f, T_f at $N = 40$ are below 10^{-220} for all three profiles. This bounds the heat values; the script's root and derivative comparisons remain finite numerical diagnostics, not a full interval-arithmetic existence proof. The intermediate-value sign bracket and the positive numerical derivative provide independent basic checks on the identified local roots.

The previous archive's verification script was inspected, extracted without changing its physics, and run to a separate output directory. Its 54 checks pass. Original file hashes, the new script, CSV results, full diagnostic JSON, environment versions, and the immediately preceding archive are included. No public repository, original PDF, or publication record was modified.

6. Meaning for T-1

The shared-action postulate accomplishes something the previous unconstrained Einstein–Maxwell action did not: for fixed spectral data it relates the gauge coefficient to gravity, and the explicitly tuned reference model supplies an isolated stationary radius in its symmetric ansatz.

It also exposes three inputs that must not be hidden. The positive trace does not cancel its own vacuum energy. The overall normalization remains free unless supplied as another physical postulate. The full profile is not determined merely by its first moments, even when all leading local heat coefficients agree.

This new action is not the old wormhole/radion theory. Its radius kinetic function, higher-derivative modes, and quantum prescription differ. The prior radion masses, lattice certificates, and broken-vacuum stability inequalities must not be imported as stability proofs for this new bosonic functional. The fermion representations are inherited, but the complete quantum theory is not established.

The action-level restriction also needs protection or a definition at the quantum level. The preceding one-loop computation showed the relevance of local kinetic data in an effective theory. A proposed spectral ultraviolet rule must state how radiative matching is performed without double-counting the spectral action as a determinant or silently reintroducing independent finite kinetic coefficients.

Finally, this calculation determines only the pure hypercharge coefficient in a fixed convention. It does not calculate the geometric weak coefficient, mixed kinetic terms, complete electroweak breaking, or running to the physical electromagnetic measurement scale. Even a fully fixed hypercharge prediction would not by itself be a derived α.

The useful next mathematical requirement is therefore sharply defined: select the actual profile and normalization from an independently justified microscopic rule and verify its quantum and vacuum consistency. Arbitrarily selecting the exponential because it is convenient is a new assumption, not completion of that requirement. The supplied change log explicitly calls for an action/measure and a physical normalization computation [P1]; this note tests one such action postulate but does not promote it to an established theory.

Sources and distinctions

[**P1**] Supplied *A Theory of Everything*, Part 1 of 4, opening claim-by-claim change log, research snapshot 19 September 2026, pages 7–11. It identifies the electromagnetic normalization bridge as open, distinguishes an illustrative compactification from the corpus's action, and calls for a physical charge/measure/scale normalization.

[**P2**] Supplied *A Theory of Everything*, Part 3 of 4, opening section-audit insert, page 1. The archived broad ToE claims do not supersede the audited current technical reference. This note does not reactivate those historical statements.

[**P3**] Saved conversation archive `t1_spectral_gauge_matching_2026-10-12.zip`, `RESEARCH_NOTE.md`, section 1, and `verify_spectral_matching.py`. Source of the inherited six-field block, four-copy count, minimally coupled sphere spectrum, and $y/2$ connection convention. The entire preceding archive is preserved unchanged in the new package.

[**1**] A. H. Chamseddine and A. Connes, *The Spectral Action Principle*, Communications in Mathematical Physics **186** (1997), 731–750, arXiv:hep-th/9606001. Source of the spectral-action concept. This note does not identify its particular ordinary sphere model with the Standard Model spectral triple studied there.

[**2**] D. V. Vassilevich, *Heat kernel expansion: user's manual*, Physics Reports **388** (2003), 279–360, arXiv:hep-th/0306138, especially the Dirac operator and the a_0, a_2, a_4 coefficients. Source of the external heat-coefficient formulas and derivative-expansion framework.

[**3**] R. Camporesi and A. Higuchi, *On the eigenfunctions of the Dirac operator on spheres and real hyperbolic spaces*, Journal of Geometry and Physics **20** (1996), 1–18, arXiv:gr-qc/9505009. Standard ordinary-sphere spectrum.

[**4**] R. Jante and B. J. Schroers, *Dirac operators on the Taub-NUT space, monopoles and SU(2) representations*, arXiv:1312.4879. Monopole Dirac and zero-mode representation input.

[**5**] A. H. Chamseddine and A. Connes, *The Uncanny Precision of the Spectral Action*, arXiv:0812.0165. Prior exact spectral-action work on different symmetric spaces.